EE445 Mod2-Lec1: Introduction to Linear Models

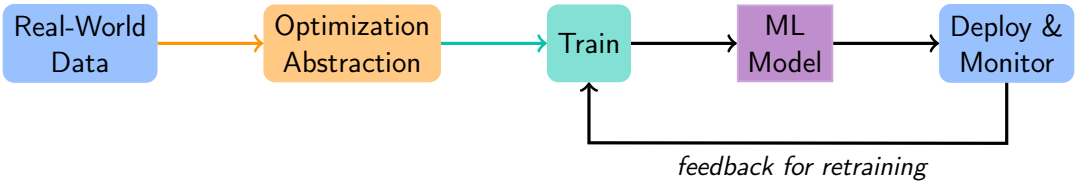
References:

- [VMLS]: Chapter 12

Agenda

- Introduction to linear models
- Least squares problem (linear regression)
- Example application: advertising purchases
- Optimality via vector calculus and linear algebra

ML workflow: Role of Linear Algebra & Optimization



This course so far:

- Optimization & Linear Algebra: Data → Optimization Abstraction via losses as norms
- ML models: Distance metric based models: k-NN, k-means

Coming up next:

- Training: Optimizing "convex" objectives (least squares)
- ML models: Linear models: linear regression, PCA, attention

ML workflow: Optimization Framework for Learning

Real world task involving data $\longrightarrow$ Optimization Problem $\longrightarrow$ ML model

Canonical Optimization Problem

$$\min_{\theta \in \Theta \subset \mathbb{R}^n} \quad \ell(\theta)$$

Terminology:

- θ: Parameter/optimization or decision variable
- $\ell(\theta)$: Objective/Loss function; in ML will often depend on data Z; write as $\ell(\theta; Z)$.
- $\Theta \subseteq \mathbb{R}^n$: Constraint set

Linear Models: Predicting Outputs from Inputs

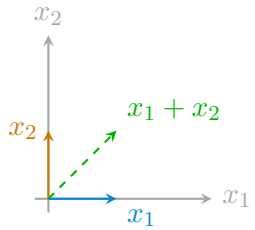
- **Goal**: learn a function (aka "model") $f_\theta(x)$ that predicts quantity $y \in \mathbb{R}$ from input $x \in \mathbb{R}^d$.
- For linear models:

- Given data (x_i, y_i) for $i = 1, \ldots, n$, we choose θ to minimize prediction error.
- Examples:
  - Predict house price from size, bedrooms, age.
  - Predict exam score from study hours.

Linearity of Functions

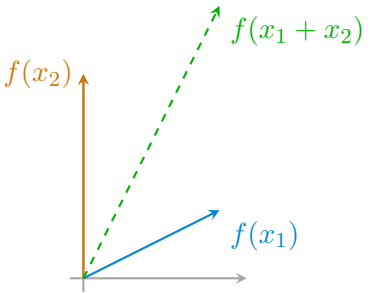
Definition: A function $f : \mathbb{R}^d \to \mathbb{R}^m$ is **linear** if for all vectors x_1, x_2 and scalars α, β:

$$f(\alpha x_1 + \beta x_2) = \alpha f(x_1) + \beta f(x_2)$$

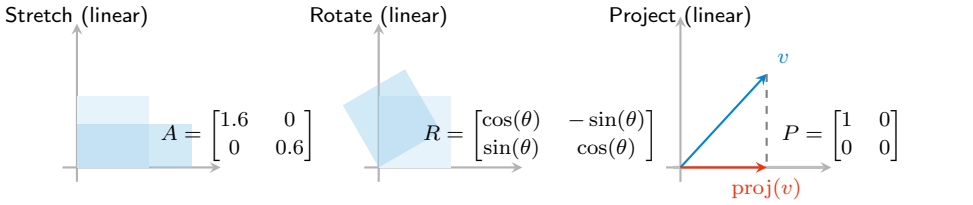


Intuition:

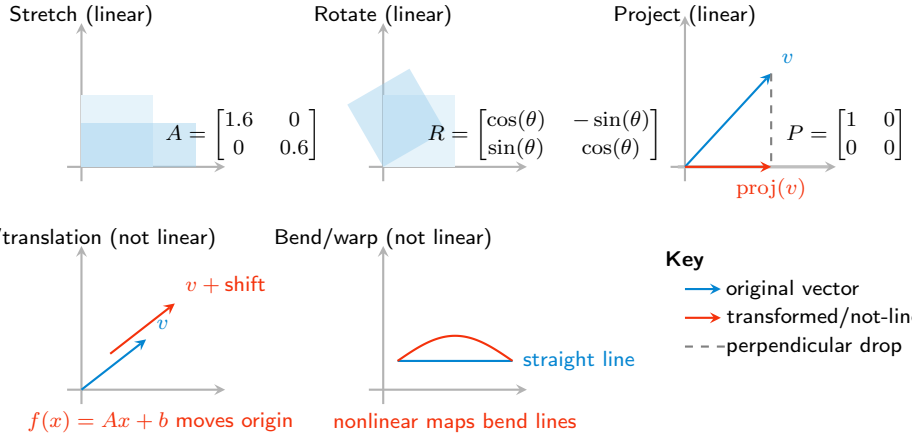
- Linear functions preserve addition and scaling.
- Geometrically, they stretch, rotate, or project vectors, but never bend or shift them.



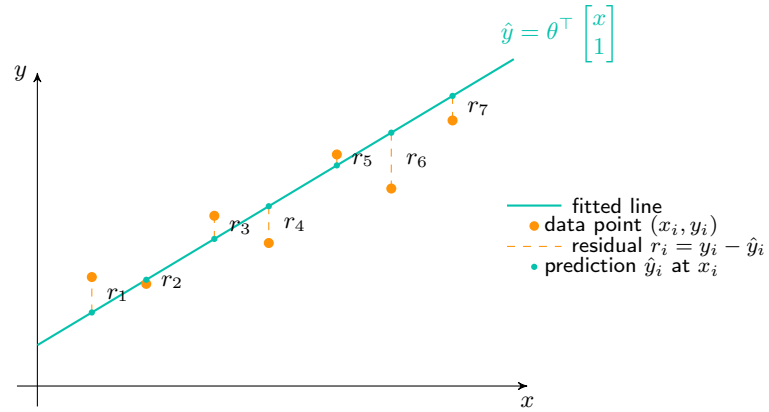
What Linear Models Can (and Cannot) Do



What Linear Models Can (and Cannot) Do



Measuring Predictor Error in 2D (Linear Regression)

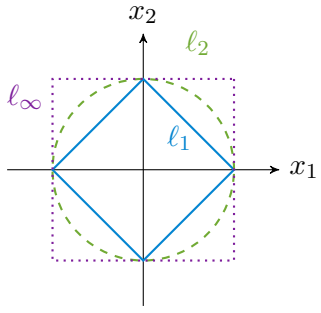


Loss Functions Measure Prediction Error

- The residual for data point i is $r_i = y_i - f_\theta(x_i)$.
- We measure total error via a *loss*:

$$\mathcal{L}(\theta) = \sum_{i=1}^{n} \|r_i\|_p$$

where $\|\cdot\|_p$ is the ℓ_p norm.



- ℓ_2 norm $\|x\|_2 = (\sum_i x_i^2)^{1/2}$: penalizes large errors strongly (smooth).
- ℓ_1 norm $\|x\|_1 = \sum_i |x_i|$: robust to outliers (sparse errors).
- ℓ_∞ norm $\|x\|_\infty = \max_i\{|x_i|\}$: minimizes worst-case error.

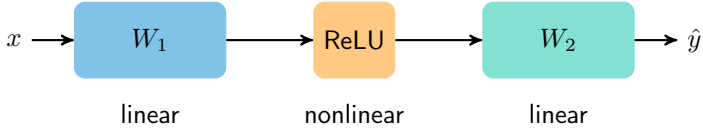
Linear Models: The Workhorse of ML

Linear Models Are Cool!

- Linear regression may look simple, but it's the foundation of modern ML.
- Every deep network layer, transformer block, and embedding lookup is built from linear maps:

$$h_{k+1} = W_k h_k + b_k \quad \text{(linear transformation)}.$$

- Nonlinearities (`ReLU`, `softmax`, etc.) just compose these linear pieces.



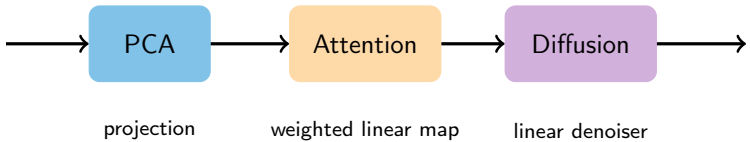
Takeaway: Modern ML = many linear models stacked + nonlinearities.

Linear Algebra is the Engine of Modern ML

- All large-scale ML systems are dominated by linear algebra:

  matrix multiplications, projections, least-squares updates.

- Examples:
  - Transformers: multi-head attention uses dot products → projections.
  - PCA: dimensionality reduction = best low-rank linear approximation.
  - Diffusion models: each denoising step fits a linear prediction to noisy data.



Message: Even billion-parameter models rely on the same math we use in least squares.

Linear Layers Are Learnable Projections

- In least squares, we *project* y onto the column space $\mathcal{C}(X)$ of X (given data).
- In neural networks, weights W define the subspace we project onto.
- Training = adjusting W so that projections align well with desired outputs.

learned projection

Same idea: find the best projection, where the subspace (weights) changes with data.

Attention Heads Are Learnable Linear Projections

Key idea: Even in modern architectures like Transformers, the core operation inside each attention head is **linear in the inputs.**

Recall: what makes a model linear?

output is *weighted sum* of its inputs: $\hat{y} = \sum_i w_i \, x_i.$

Note: the weights can depend on context, but the combination itself is linear.

Attention does exactly this!

Each attention head: Output for item $i = \sum_j$ (weight from i to j)$\times$(embedded input j).

- The model learns how to choose the weights (who pays attention to whom).
- Once the weights are set, the output is a linear combination of the inputs.

A Simple View: Linear Map on Learned Embeddings

Key idea: Once an attention head is trained, its weights are fixed. At inference, it performs a sequence of **linear transformations** on the input embeddings.

$$\text{output} = W\, f(X)$$

Interpretation:

- X: the input embedding (learned representation of the data)
- $f(X)$: a transformation that mixes or reweights embeddings based on context
- W: a fixed linear map learned during training

Big idea: Even in complex architectures like attention heads, the computation reduces to a *linear transformation* of the current embedding.

We addressed "why linear", so now why start with least squares?

Why Start with Least Squares?

- It's the simplest form of empirical risk minimization:

- It teaches:
  - Geometry of learning (projection, subspaces)
  - Optimization (gradients, convexity)
  - Regularization (bias–variance, ridge regression)
- These same ideas reappear in:
  - Linear classifiers (SVMs, logistic regression)
  - Neural networks (gradient descent on squared loss)
  - Reinforcement learning (least-squares temporal difference)

Once you understand least squares, you understand the logic behind all of these.

Empirical Risk Minimization (ERM): Quientissential ML Problem

Goal of supervised learning: Find a function that makes good predictions on new data.

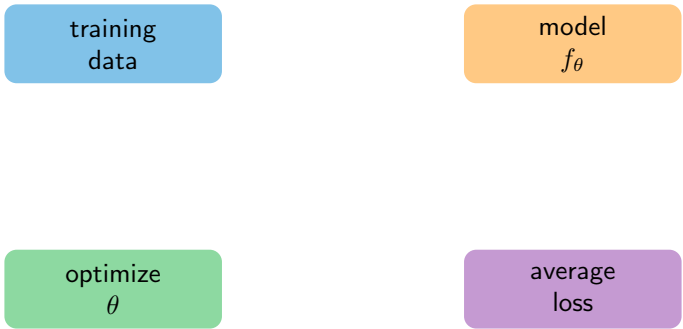
Challenge: We don't know the true data distribution; indeed, we only see a finite set of samples.

Idea of ERM:

Why it matters: ERM formalizes the idea of *fitting a model to data*.

- Almost every ML algorithm—from linear regression to deep networks—minimizes some form of empirical risk.
- Least squares is the simplest example.

ERM: The Big Picture



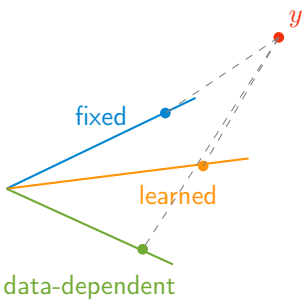
Takeaway: ERM formalizes model training as minimizing the average loss—the *empirical risk*—and the process repeats as new data arrive.

Big Picture: ML as Learning to Project

ML algorithms are understood as **projections onto subspaces**—some fixed, some learned. That is, they find the closest point in a subspace to the data.

Method	**Projection target (subspace)**	**How it's chosen**
Least Squares	Columns of X (given data)	Fixed by features
PCA	Top eigenvectors of $X^\top X$	Learned to maximize variance
Attention	Value vectors V weighted by similarity of q and K	Learned, data-dependent
Diffusion Models	Linear prediction of noise component	Learned through training

Big Picture: ML as Learning to Project



Color	Represents	Meaning in ML Context
■	**Least Squares** subspace	Fixed column space of X: the features you choose define the projection.
■	**PCA** subspace	Low-dimensional directions learned from data to maximize variance.
■	**Attention** subspace	Data-dependent subspace; each query q changes the projection space dynamically.

Core insight: All these methods find or use a subspace where predictions are "closest" to data; the difference is *how the subspace itself is learned*.

Least Squares: The Simplest Form of ERM & Supervised Learning

Least Squares Set-up

- Fitting a “linear regression model" is the simplest form of machine learning out there.
- Consider an $n \times d$ matrix $X \in \mathbb{R}^{n \times d}$ and vectors $y \in \mathbb{R}^n$, $\theta \in \mathbb{R}^d$
- **Goal**: find θ such that $X\theta = y$
- Intepretation:
  - training data $X \in \mathbb{R}^{n \times d}$; e.g., n is number of samples, d is number of ‘features’
  - $y \in \mathbb{R}^n$ contains ‘target values’, ‘labels’, or observations
  - $\theta \in \mathbb{R}^d$ is a set of feature weights

Example Application: Advertising Purchases

- Consider n demographic groups that we want to advertise to with a target # 'impressions' y for each group
- We purchase advertising in d different channels (e.g., different web publishers, radio, print,...), in amounts given by $\theta \in \mathbb{R}^d$.
- Entry x_{ij} in matrix $X \in \mathbb{R}^{n \times d}$: # impressions in group i per dollar spent on advertising in channel j.
  - jth column of X gives the effectiveness or reach (in impressions per dollar) for channel j.
  - ith row of X shows which media demographic group i is exposed to.

- **Goal**: find θ such that $\|X\theta - y\|_2^2$ is as small as possible (minimized)

Overdetermined System of Equations→Least Squares Opt

- **Goal**: Find a solution to $X\theta = y$—that is, find θ such that $X\theta = y$
- typically, X is a 'tall' matrix (so we get an 'over-determined' system); there are more equations (n) than variables to choose (d).

- there's often no exact solution → formulate an **optimization problem** to find as *close* a solution as possible; i.e., an *least squares approximate solution*

Least Squares Formulation

Least Squares Formulation: Empirical Risk Minimization

- $X \in \mathbb{R}^{n \times d}$: the *design matrix*, rows are data points $x_i^\top$.
- $y \in \mathbb{R}^n$: vector of targets.
- ℓ_2 loss is smooth, convex, and differentiable $\rightarrow$ analytic solution exists.

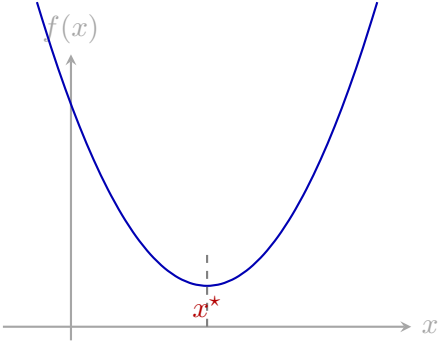
Finding Minima and Convexity

Goal: Find $\theta^\star$ that minimizes a differentiable function $f : \mathbb{R}^d \to \mathbb{R}$.

Conditions for Minima

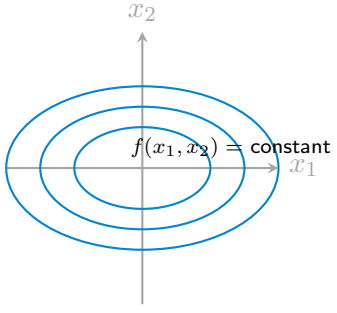
Finding Minima in One Dimension

For a smooth function $f : \mathbb{R} \to \mathbb{R}$, a minimum occurs when:



Minima in Two Dimensions

Now $f : \mathbb{R}^2 \to \mathbb{R}$ has a gradient and Hessian:



General Convexity Condition via Eigenvalues

For any symmetric Hessian $H \in \mathbb{R}^{d \times d}$,

$$H \text{ is positive definite} \iff v^\top H v > 0 \text{ for all } v \neq 0.$$

Checking this directly is hard, so we use **eigenvalues**.

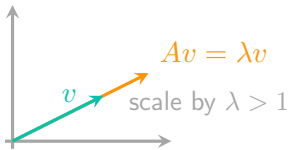
Eigenvalues and Eigenvectors

For a square matrix $A \in \mathbb{R}^{d \times d}$:

Geometric meaning:

- A scales (stretches or flips) v without changing its direction.
- The eigenvalues describe how much A stretches space along special directions.
- If $\lambda = 0$, that direction is *flattened* $\rightarrow$ indicates "rank deficiency".

Rank and eigenvalues:



Why Eigenvalues Indicate Positive Definiteness

Let $A \in \mathbb{R}^{d \times d}$ and consider the quadratic form:

Interpretation:

- Each eigenvector v defines a special direction in space.
- Along that direction, A scales by λ and stretches by λ^2 in $A^\top A$.
- If $\lambda^2 > 0$ for all eigenvalues, then $A^\top A$ is **positive definite** (i.e. $A^\top A \succ 0$).

Therefore:

What are we missing?

- First, we have $v^\top H v$, but we just looked at a condition on matrices like $A^\top A$. So can we write H as $A^\top A$ for some A?
  - Yes! If H is symmetric and positive semidefinite (i.e. $H \succeq 0 \iff x^\top H x \geq 0,\ \forall\, x$).
  - We know its symmetric because of its structure as a Hessian ($H = H^\top$)
- What is linear independence of eigenvector?

Independence of Eigenvectors and Basis Formation

Linear Independence

Independence of Eigenvectors and Basis Formation

Theorem

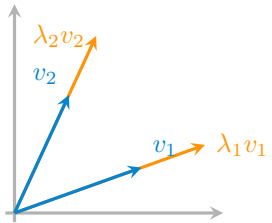
If $Av_i = \lambda_i v_i$ and $Av_j = \lambda_j v_j$ with $\lambda_i \neq \lambda_j$, then v_i and v_j are linearly independent.

Proof sketch:

Independence of Eigenvectors and Basis Formation

Facts

- For $A \in \mathbb{R}^{d \times d}$ with d distinct eigenvalues, its eigenvectors $v_1, \ldots, v_d$ are linearly independent.
- If A has d distinct eigenvalues, its eigenvectors form a basis for $\mathbb{R}^d$.
- Distinct eigenvalues guarantee independent eigenvectors, which form a basis that diagonalizes A.

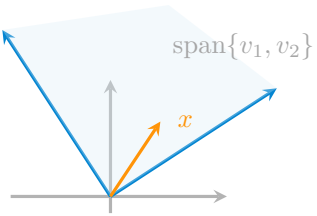


What is a Basis?

Definition of Basis

Key Ideas:

- "Span" $\implies$ the basis vectors *reach* all of V.
- "Linear independence" $\implies$ no vector in the set is redundant.
- The number of basis vectors = **dimension** of V.



Independence of Eigenvectors and Basis Formation

Why this matters:

Interpretation:

- Eigenvectors give A's *natural coordinate system*.
- Their independence guarantees we can describe any vector as

$$x = \alpha_1 v_1 + \cdots + \alpha_n v_d.$$

- A then acts by simply scaling each coordinate by λ_i.

Where are we

- We have defined linear independence, and even shown how to diagonalize a matrix if it has independent eigenvectors.
- Now we want to come back to this Hessian matrix and questions about curvature.
- To apply the eigenvalue check, we need to show that Hessians can be decomposed as $H = A^\top A$ for some A.

Why Symmetric Matrices Can Be Diagonalized

Claim (Spectral Theorem)

Intuition:

- Symmetry $\Rightarrow$ all eigenvalues are **real**.
- Symmetry $\Rightarrow$ eigenvectors are **orthogonal** (i.e., $\langle v_i, v_j \rangle = 0$ for $i \neq j$).
- Then $Q^\top Q = I$ so that $HQ = Q\Lambda \implies H = Q\Lambda Q^\top$.

Checking Convexity with the Hessian

$$H(\theta) = \nabla^2 f(\theta) = \begin{bmatrix} \frac{\partial^2 f}{\partial \theta_1^2} & \cdots & \frac{\partial^2 f}{\partial \theta_1 \partial \theta_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial \theta_d \partial \theta_1} & \cdots & \frac{\partial^2 f}{\partial \theta_d^2} \end{bmatrix} \qquad \text{(Hessian)}$$

$H(\theta) \succ 0 \iff v^\top H(\theta) v > 0 \quad \forall\, v \neq 0 \implies f$ is **strictly convex** at that point

Eigenvalue test:

Eigenvalues and Curvature: PD, PSD, and Indefinite

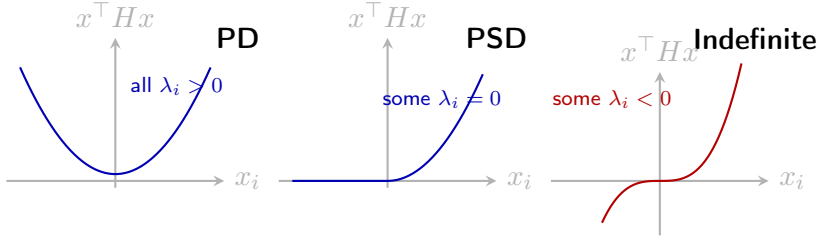
For a symmetric matrix H, curvature along direction v is measured by

$$v^\top H v = \lambda \|v\|_2^2.$$

$\lambda_i > 0 \Rightarrow$ curves upward (bowl)

$\lambda_i = 0 \Rightarrow$ flat direction (ridge)

$\lambda_i < 0 \Rightarrow$ curves downward (saddle)



Takeaway:

Summary: Positive Definiteness, Convexity, and Optimality

- Optimality: $\nabla f(\theta^\star) = 0$ and $H(\theta^\star) \succ 0 \implies \theta^\star$ is a local minimum.
- Convexity: $H(\theta) \succ 0$ for all $\theta \implies f$ is convex $\implies$ any stationary point is a global minimum.
- Positive definiteness: $H \succ 0 \iff$ all eigenvalues $\lambda_i > 0$.
  - 1D: $f''(x) > 0$.
  - 2D: both diagonal entries > 0.
  - General d: all eigenvalues of H positive $\implies$ convex function.

Solving Least Squares (Coordinate Form)

For one data point: $f_\theta(x_i) = \theta_1 x_{i1} + \theta_2 x_{i2} + \cdots + \theta_d x_{id} = \theta^\top x$.

$$\text{minimize } \mathcal{L}(\theta) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \theta_1 x_{i1} - \theta_2 x_{i2} - \cdots - \theta_d x_{id})^2.$$

Take partial derivatives with respect to each θ_j:

Solving Least Squares (Vector Form)

$$\mathcal{L}(\theta) = \frac{1}{n}\|y - X\theta\|_2^2 = \frac{1}{n}(y - X\theta)^\top (y - X\theta).$$

Coordinate vs. Vector Gradient (2D Example)

Consider two features: $x_i = [x_{i1},\, x_{i2}]^\top$ and parameters $\theta = [\theta_1,\, \theta_2]^\top$.

$$\mathcal{L}(\theta) = \frac{1}{n}\sum_{i=1}^{n}(y_i - x_{i1}\theta_1 - x_{i2}\theta_2)^2 = \frac{1}{n}\|y - X\theta\|_2^2.$$

Coordinate form:

$$\frac{\partial \mathcal{L}}{\partial \theta_1} = -\frac{2}{n}\sum_{i=1}^{n} x_{i1}(y_i - x_{i1}\theta_1 - x_{i2}\theta_2),$$

$$\frac{\partial \mathcal{L}}{\partial \theta_2} = -\frac{2}{n}\sum_{i=1}^{n} x_{i2}(y_i - x_{i1}\theta_1 - x_{i2}\theta_2).$$

Vector form:

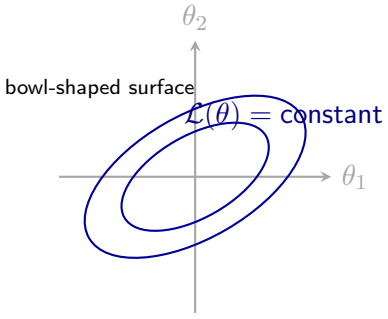
$$\nabla_\theta \mathcal{L}(\theta) = -\frac{2}{n}X^\top(y - X\theta)$$

$$= -\frac{2}{n}\begin{bmatrix} \sum_i x_{i1}(y_i - x_{i1}\theta_1 - x_{i2}\theta_2) \\ \sum_i x_{i2}(y_i - x_{i1}\theta_1 - x_{i2}\theta_2) \end{bmatrix}.$$

Uniqueness: Convexity of the Least Squares Objective

$$\min_{\theta} \mathcal{L}(\theta) = \frac{1}{n}\|y - X\theta\|_2^2 = \frac{1}{n}(y - X\theta)^\top (y - X\theta) \implies \theta^\star = (X^\top X)^{-1} X^\top y.$$

Convexity Intuition (2D Example)



Takeaway: Least squares is convex because its Hessian $\frac{2}{n}X^\top X$ is positive definite when X has full column rank.

What happens if some directions have zero curvature?

Next time

Today we covered:

- Convexity, Hessians, and curvature
- Linear independence, basis, and dimension
- Why symmetric matrices can be diagonalized
- Checking convexity with the eigenvalue test
- Solving least squares by setting the gradient to zero

Next time we will cover:

- Linear Algebra Concepts: Matrix Rank and the Pseudoinverse, Under- vs Over-determined systems
- Projections onto subspaces and connections to least squares
- Meaning/interpretation of least squares as a ML model